# Anubhav Kulshreshtha

kulshreshtha.anubhav2003@gmail.com | +91 6398256821 | Portfolio | Codolio
linkedin.com/in/anubhav-kulshreshtha | github.com/anubhav1450

## PROFILE

Backend-focused software engineer (B.Tech CSE, 2026) with internship experience building REST APIs, authentication and database-backed workflows in Node.js. Built a real-time collaborative editor on Yjs CRDTs and a Python system that puts an LLM behind signed-token authorization, backed by 86 automated tests. Top 7.63% on LeetCode with an 1817 peak contest rating.

## SKILLS

**Languages:** Python, JavaScript, Java, SQL

**Software Engineering:** Data Structures & Algorithms, OOP, SDLC, Unit & API Testing, Debugging

**Backend:** Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT Authentication, RBAC

**Databases:** MongoDB, MySQL, SQLite

**Cloud & Tools:** AWS ECS, Docker, Vercel, Render, Git, Linux, Postman, Playwright, LLM API integration

**Frontend:** React.js, HTML5, CSS3, Tailwind CSS

## PROFESSIONAL EXPERIENCE

**Lighthouse Info Systems Pvt Ltd,** *Software Engineer Intern* May 2025 – July 2025 | Nagpur, India

- Built **backend modules** for product management, cart operations and order processing in Node.js, Express.js and MongoDB, implementing the end-to-end order workflow.
- Designed **REST APIs** with **JWT authentication** and role-based access control (RBAC) over structured MongoDB schemas.
- Added **input validation, structured error handling and API tests** across endpoints for consistent behavior.

## PROJECTS

**Real Time Collaborative Editor**

- Built a **real-time collaborative code editor** with React, Monaco Editor and **Yjs CRDTs** synced over Socket.io on a Node.js server, so simultaneous edits from many users merge without conflicts.
- Added **room-based sessions with live presence**: shareable room codes and invite links, an active-users panel, colored remote cursors with name tags, and a language setting synced across the room.
- Verified sync and presence with a **Playwright test running 20 concurrent browser sessions** in one room; containerized with Docker (multi-stage build, health checks) and deployed on Vercel and Render.
- **Technologies:** React.js, Node.js, Express.js, Socket.io, Yjs, Monaco Editor, Tailwind CSS, Docker, Playwright

**Agent Trust Lab**

- Built a multi-agent refund workflow in Python where the **Payment API verifies HMAC-signed, limit-bound delegation tokens** before acting, blocking over-limit requests and forged tokens regardless of what the agent decided.
- Integrated a **real LLM (Groq, OpenAI-compatible API)** as an untrusted decision-maker: its JSON output is validated and **fails closed** when invalid, and responses are recorded by request hash so an unchanged replay makes **zero model calls**.
- Logged every step in a **hash-chained, tamper-evident event log** with sandboxed replay to test which input caused an outcome; covered by **86 automated tests** and deployed as a FastAPI + SQLite service with a live dashboard.
- **Technologies:** Python, FastAPI, SQLite, HMAC / SHA-256, LLM API (Groq), JavaScript, Vercel

## EDUCATION

**B.Tech. in Computer Science and Engineering** 2022 – 2026 | Bharatpur, India
*Government Engineering College Bharatpur, CGPA: 8.65*

## ACHIEVEMENTS AND ACTIVITIES

- **Top 7.63% on LeetCode** with an **1817 Peak Contest Rating**; solved **600+ DSA problems** across LeetCode, CodeChef, and GeeksforGeeks.
- Served as **Coordinator of the Python Programming Club**, organizing technical workshops, coding events, and a hackathon for **50+ students**.
- Earned **NPTEL** Data Structures and Algorithms with Java certification (IIT-backed).